CSE 752 Software Architecture and Design Practices

Architecture Drivers
(50 marks)

Objective of activity: To systematically develop architecture drivers for a system

Outcomes: Be able identify architecture drivers from functional architecture

Stakeholders

S. No.	Stakeholder name	Description of their role in the system
1	Doctors	They provide the primary treatment.
2	Patients	They are the owners of the cards.
3	Hospital Admin	They manage the hospital resources that other entities engage in.
4	Nurses	They record vitals, support consultations and assist with patient care and follow-up.
5	Lab technicians and pharmacists	They process test orders and samples, and dispense e-prescriptions.
6	Front-desk / registration staff	They register patients, issue or replace cards and manage check-in and appointments.
7	Insurers	They receive itemised bills and settle claims through the insurer interface.
8	Third-party developers and facility IT	They build applications and integrations on the open APIs, SDKs and specifications.
9	Registry / governing authority	It operates the shared identity, consent, audit and registry services that all facilities rely on.

Functional Features

1. (10 marks) Group the entire functionality into a list of no more than 20 functional features of your system. Use the following template for the response:

Feature_ID	Feature Name	Invoking stakeholder	Feature Description (1 sentence)
F1	Universal Patient Identity	Patient, Front-desk staff	Issues one persistent, non-meaningful UPID per patient and resolves duplicate or conflicting identities across facilities.
F2	Authentication & Card Credentials	Patient, Doctors, Staff	Authenticates patients and staff and binds QR/RFID/NFC card tokens to a UPID, with instant revocation and vendor-neutral reader adapters.

F3	Consent & Audit Trust	Patient, Registry authority	Evaluates every access request against the patient's consent policy and records it in a tamper-evident audit log.
F4	Interoperability Gateway & Record Locator	Facilities, Doctors	Provides a standards-based API layer and an index of which facility holds which category of record for a UPID.
F5	Shared Registries & Standards	Hospital Admin, Registry authority	Maintains a verified directory of facilities and providers and the agreed code sets and schemas used for data exchange.
F6	Open Developer Toolkit & Offline Sync	Third-party developers, Facility IT	Offers open APIs, SDKs, documentation, a reference implementation and an offline sync agent for applications.
F7	Registration & Appointments	Front-desk staff, Patient	Registers patients, issues or replaces cards and handles appointment slots, booking and check-in.
F8	Clinical Care Suite	Doctors, Nurses, Lab technicians, Pharmacists	Supports consultation, diagnosis, orders, laboratory work and e-prescription with pharmacy dispensing.
F9	Patient Engagement & Follow-up	Patient, Doctors	Gives patients a dashboard, sends SMS/email/push notifications and suggests follow-up actions based on rules.
F10	Billing & Administration	Hospital Admin, Insurers	Produces itemised bills and insurer claims and manages staff roles and facility configuration.

Functional Modules

2. (10 marks) Group functionality into no more than 6 functional modules

Module_ID	Functional Module Name	Stakeholder Category	Brief description
M1	Identity & Access	Patients, front-desk staff, doctors, registry authority	Establishes who a person is and how they prove it. Issues one opaque UPID per patient, resolves duplicate identities with human review, authenticates patients and staff with short-lived tokens, and binds QR/RFID/NFC cards to a UPID with instant revocation. Uses a vendor-neutral card-reader interface. Called at run time by every application (M4, M5, M6) and by M2.
M2	Trust & Data Exchange	Registry authority, patients, facilities, doctors	Controls who may see what and moves data between facilities. Evaluates each request against the patient's consent policy, routes it

			through a standards-based gateway, finds records through a locator that holds pointers only, and writes a tamper-evident audit log. Maintains the facility/provider directory and shared code sets and schemas. Depends on M1 for identity; called at run time by M5 and M6.
M3	Open Ecosystem	Third-party developers, facility IT, vendors	Supplies the open specifications, APIs, SDKs, documentation, reference implementation and card-reader specification that M1 and M2 are defined by and that M4, M5 and M6 are built with. Includes the offline sync agent (local cache, operation queue, resync, conflict flagging), which is embedded in the applications. Mostly build-time; only the sync agent runs inside applications.
M4	Access to Care	Patients, front-desk staff, hospital admin	The front-desk application. Registers patients against a UPID, issues, activates and replaces cards (blocking lost ones), and manages appointment slots, booking, rescheduling and card-based check-in. Built with M3; calls M1 at run time for identity and cards.
M5	Clinical Care	Doctors, nurses, lab technicians, pharmacists	The clinical application. Provides the doctor and nurse workspace for vitals, consultation and diagnosis, handles lab orders, samples and results, and supports e-prescription and pharmacy dispensing. Keeps detailed records at the facility and pulls permitted history from other facilities. Built with M3; calls M1 and M2 at run time.
M6	Engagement & Operations	Patients, hospital admin, insurers, doctors	The patient-facing and back-office application. Gives patients a dashboard, suggests rule-based follow-ups, sends SMS/email/push notifications, produces itemised bills and the insurer interface, and provides the administration console for staff roles and facility configuration. Built with M3; calls M1 and M2 at run time.

Zachman Framework Row #1 + #2 (combined)

3. (10 marks) Create a **VIEW** that contains the columns of the Zachman Framework from the **VIEWPOINT** of the first two rows (combined) and the **security** and **scalability** PERSPECTIVE.

System Owner View for Functional Module #1: Identity & Access

Zachman F/W Column	Identity & Access
Why	Give every patient one trusted identity and a secure way to prove it at any registered facility.
What	UPID, identity records and match decisions, login tokens, card credentials (QR/RFID/NFC) and their status.
Who	Patients (card owners), front-desk staff, doctors, registry authority.
How	Issue a UPID once, match and merge duplicates with human review, authenticate users and bind or revoke cards.
When	At first registration, at every visit or login, on card loss or replacement, and when duplicates are found.
Where	Shared national/regional identity service used by all facilities; card readers at facility entry points.

Perspectives for Functional Module #1: Identity & Access

Perspective	Identity & Access
Security	UPID carries no personal data; short-lived tokens; immediate card revocation visible everywhere; human review before any merge.
Scalability	One shared namespace that must serve all facilities and millions of patients; matching and authentication must scale horizontally.

System Owner View for Functional Module #2: Trust & Data Exchange

Zachman F/W Column	Trust & Data Exchange
Why	Ensure patient data is shared only with consent and that every access can be proven.
What	Consent policies, access decisions, audit events, record-location pointers, facility/provider registry, code sets and schemas.
Who	Patients (grant consent), registry authority, doctors and facilities requesting data.
How	Evaluate consent on each request, log it append-only, locate records by pointer and exchange data through a standards-based gateway.

When	On every cross-facility data request, when consent changes, and when facilities or schemas are added or updated.
Where	Shared services outside any one facility; clinical data itself stays at the originating facility.

Perspectives for Functional Module #2: Trust & Data Exchange

Perspective	Trust & Data Exchange
Security	Consent enforced identically at every facility; tamper-evident audit under independent control; locator holds no clinical content; certificate-based facility trust.
Scalability	Gateway needs rate limiting and routing for many facilities at once; locator and registries must handle growing lookups.

System Owner View for Functional Module #3: Open Ecosystem

Zachman F/W Column	Open Ecosystem
Why	Let anyone implement and extend the system without vendor lock-in.
What	Open APIs, SDKs, schemas, documentation, reference implementation, card-reader specification and offline sync agent.
Who	Third-party developers, facility IT teams, vendors, pilot and training teams.
How	Publish open specifications and code, provide a deployable reference stack and embed an offline agent in applications.
When	During pilots and onboarding, on each new release and whenever facilities work with poor connectivity.
Where	Public repositories and documentation; the sync agent runs inside applications at each facility.

Perspectives for Functional Module #3: Open Ecosystem

Perspective	Open Ecosystem
Security	Open specs must not weaken access control; sync agent holds minimal local data with queued operations protected on the device.
Scalability	Many independent implementers and versions must interoperate; offline queues must sync and resolve conflicts at scale.

System Owner View for Functional Module #4: Access to Care

Zachman F/W Column	Access to Care
Why	Get patients registered, identified and booked quickly at the front desk.
What	Registration forms, card issue/replacement records, appointment slots, bookings and check-in status.
Who	Front-desk staff, patients, hospital admin.
How	Register against a UPID, issue or replace cards, book slots and check patients in by scanning their card.
When	At first visit, at card loss or replacement, when booking and on the day of the appointment.
Where	Facility front desk, registration counters and patient-facing booking channels.

Perspectives for Functional Module #4: Access to Care

Perspective	Access to Care
Security	Verify identity before issuing or replacing a card; staff actions logged; only authorised staff can register patients.
Scalability	Peak-hour check-in and booking load per facility; app must work as the number of facilities and patients grows.

System Owner View for Functional Module #5: Clinical Care

Zachman F/W Column	Clinical Care
Why	Support clinicians in delivering safe, well-informed treatment.
What	Vitals, consultation notes, diagnoses, lab orders, samples and results, prescriptions and dispensing records.
Who	Doctors, nurses, lab technicians, pharmacists.
How	Pull permitted history, record the consultation, order tests, issue e-prescriptions and dispense them.
When	During each consultation, when tests are ordered or resulted and when prescriptions are issued or dispensed.
Where	Clinical workspaces, laboratory and pharmacy within the facility; summary shared outward.

Perspectives for Functional Module #5: Clinical Care

Perspective	Clinical Care
Security	Role-based access; patient records stored at the facility with only a summary shared; orders and prescriptions attributable to the clinician.
Scalability	Must stay responsive as visit volume grows and keep working when the facility's link to shared services is down.

System Owner View for Functional Module #6: Engagement & Operations

Zachman F/W Column	Engagement & Operations
Why	Keep patients informed and facilities running and funded.
What	Patient dashboard, follow-up recommendations, notification templates and logs, itemised bills, claims, staff roles and facility configuration.
Who	Patients, hospital admin, insurers, doctors.
How	Show patient views, trigger rule-based follow-ups, send SMS/email/push, generate bills and claims and manage roles.
When	After visits for follow-up, on billing events, on schedule for reminders and when administrators change configuration.
Where	Patient-facing app and channels; billing and administration at the facility; shared notification service.

Perspectives for Functional Module #6: Engagement & Operations

Perspective	Engagement & Operations
Security	Patients see only their own data; notifications avoid sensitive clinical detail; billing data stays at the facility; role-based admin.
Scalability	Notification volume spikes and growing patient base; each facility's billing and admin must scale independently.

Feature Module Mapping

4. (10 marks) Map the 20 functional features given above to the appropriate functional modules

S.No.	Feature_ID	Functional Module
1	F1 - Universal Patient Identity	M1 - Identity & Access
2	F2 - Authentication & Card Credentials	M1 - Identity & Access
3	F3 - Consent & Audit Trust	M2 - Trust & Data Exchange
4	F4 - Interoperability Gateway & Record Locator	M2 - Trust & Data Exchange
5	F5 - Shared Registries & Standards	M2 - Trust & Data Exchange
6	F6 - Open Developer Toolkit & Offline Sync	M3 - Open Ecosystem
7	F7 - Registration & Appointments	M4 - Access to Care
8	F8 - Clinical Care Suite	M5 - Clinical Care
9	F9 - Patient Engagement & Follow-up	M6 - Engagement & Operations
10	F10 - Billing & Administration	M6 - Engagement & Operations

5. (10 marks) Provide the list of architecture drivers for each functional module. Write at least one architecture driver for each functional module. More the better!

Architecture Drivers			
S.No.	Functional Module Name	Key Architecture Driver	Driver Category*
1	M1 - Identity & Access	Every patient must have exactly one persistent UPID, with duplicates detected and merged only after human review.	Concern
2	M1 - Identity & Access	The UPID must be opaque and carry no personal information.	Constraint
3	M1 - Identity & Access	Card reader integration must be vendor-neutral (QR, RFID, NFC, manual entry) with no lock-in to one vendor.	Constraint
4	M1 - Identity & Access	Revoking a lost or stolen card must take effect across all facilities at once.	Quality attribute

5	M2 - Trust & Data Exchange	Every record access must be decided by the patient's consent policy, applied identically at every facility.	Concern
6	M2 - Trust & Data Exchange	Clinical data must stay at the originating facility; the locator may hold pointers only.	Constraint
7	M2 - Trust & Data Exchange	Audit logs must be append-only, tamper-evident and controlled independently of any facility.	Quality attribute
8	M2 - Trust & Data Exchange	Exchange must follow open, FHIR-aligned standards and reuse published code sets.	Constraint
9	M2 - Trust & Data Exchange	The gateway must enforce rate limiting and scale to many facilities concurrently.	Quality attribute
10	M3 - Open Ecosystem	Specifications, SDKs and the reference implementation must be openly available to any implementer.	Constraint
11	M3 - Open Ecosystem	Facilities with poor connectivity must keep working offline and sync later with conflict flagging.	Concern
12	M3 - Open Ecosystem	Independent implementations must interoperate, with versioned APIs that stay backward compatible.	Quality attribute
13	M4 - Access to Care	Card issue and replacement must verify the patient's identity against the UPID.	Concern
14	M4 - Access to Care	Appointment booking must prevent double-booking of slots and resources.	Concern
15	M4 - Access to Care	Check-in must stay fast during peak hours.	Quality attribute
16	M5 - Clinical Care	Clinicians must be able to see permitted patient history from other facilities during a consultation.	Concern
17	M5 - Clinical Care	Detailed clinical data must be stored at the facility, with only a summary shared elsewhere.	Constraint

18	M5 - Clinical Care	Orders and prescriptions must be accurate and attributable to the issuing clinician.	Quality attribute
19	M5 - Clinical Care	Clinical workspaces must remain usable when the connection to shared services is lost.	Quality attribute
20	M6 - Engagement & Operations	Patients must be able to view only their own data through the dashboard.	Concern
21	M6 - Engagement & Operations	Notifications (SMS, email, push) must not expose sensitive clinical details.	Constraint
22	M6 - Engagement & Operations	Billing data must stay at the facility, and staff roles and permissions must be configurable per facility.	Constraint
23	M6 - Engagement & Operations	The system must handle growth in patients and spikes in notification volume.	Quality attribute

* Driver Category: Choose your architecture drivers from the following categories:

- **Concerns** that should be addressed / taken care of (What the system SHOULD do)
- **Constraints** within which system should operate (What the system SHOULD NOT do)
- **Quality attributes** of the operating environment

What to upload:

- Upload your response in the form of a PDF document
- Name of the file: <<group>>-ArchDrivers.pdf (e.g., CSE752-2025-01-ArchDrivers.pdf)